Deep Survival Prediction for mCRPC Using Multi-Trial Data

Javier Mencia

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HIVE Lab
Health Informatics, Visualization, and Equity

Prostate Cancer: Global Burden

- **2nd common cancer** in men worldwide with **1.4 million new cases** annually
- **5th leading cause** of cancer deaths in men (400k annually) [1]
- Aging populations driving increasing global burden: Incidence expected to double by 2040 [2]

Disease Progression

- Majority diagnosed with localized disease (5-year survival $> 91\%$)
- 10-20% develop metastatic disease

metastatic Castration Resistant Prostate Cancer (mCRPC): Poor Prognosis Phase

- **Castration-resistant** progression despite testosterone suppression
- **Metastatic spread** predominantly to bones (90%) and lymph nodes
- **Median overall survival**: 1-2 years
- **5-year survival rate** $< 30\%$ for metastatic prostate cancer

Treatment Challenges

- Heterogeneous response to available therapies
- Limited predictive biomarkers for treatment selection
- Significant symptom burden affecting quality of life
- High economic burden on healthcare systems

Study Objectives

Primary Aim

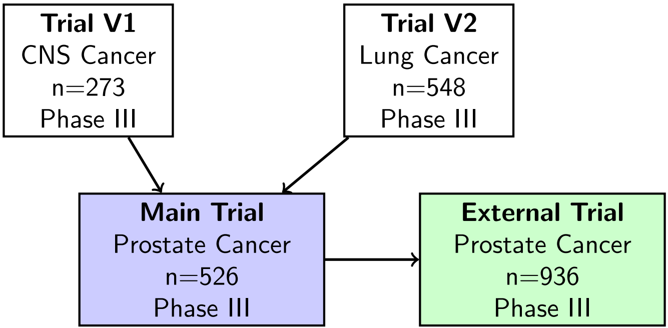
Develop and externally validate an explainable survival prediction model for mCRPC patients using multi-trial clinical data

Secondary Objectives

- Identify key prognostic factors from baseline characteristics
- Compare multiple machine learning approaches for survival analysis
- Assess clinical utility through decision curve analysis
- Provide model interpretability using SHAP and partial dependence

Data Sources - Trial Overview

Initial Validation



Initial Validation Trials

- Inform variable selection
- Assess data quality patterns
- Preliminary model testing

External Validation

- Final model performance
- Generalizability assessment
- Clinical applicability

Trials from Open Data Sphere. CNS: Central Nervous System

Main Trial - Development Cohort

- Multicenter, international, phase III trial initiated in 2009
- Randomized, double-blind, placebo-controlled design

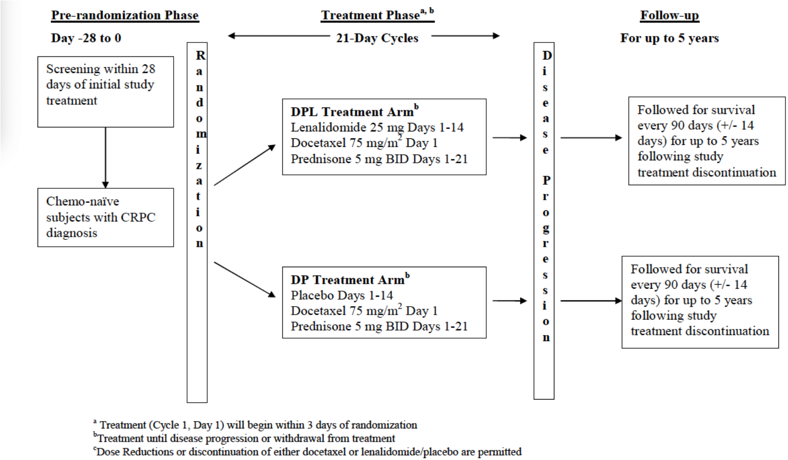


Figure: Study Schema taken from Trial Protocol

Main Trial Cohort

Table: Baseline Characteristics (N=526)

Characteristic	Value
Sex	
Male	100%
Race	
White	82.3%
Age Group	
60-69	42.8%
70+	39.0%
Region	
Western Europe	47.0%
North America	26.4%
Eastern Europe	16.0%
Treatment Exposure	
Treatment Discontinuation	52.9%
Completed 7+ cycles	62.6%

Note: Discontinuation includes patients who died (14.4%), were lost to follow-up (0.2%), or had status not reported (57.0%).

Modeling Approaches

Penalized Cox

- Restricted cubic splines
- Elastic Net Regularization

Random Survival Forests

- Ensemble method
- Handles interactions
- Non-linear effects

Deep Survival Network

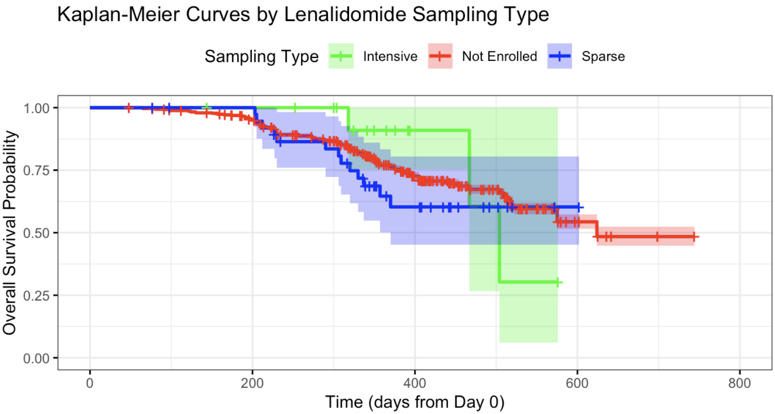
- Neural network architecture
- Flexible feature learning
- Compact design

Survival Overview

This study employed two pharmacokinetic sampling approaches:

- **Intensive PK**: 6 samples on Cycle 1 Day 1 at clinic (1, 1.5, 2, 3, 4, 6-8 hours)
- **Sparse PK**: 2 samples on Cycle 1 Day 14 at home (4 hours apart)

Intensive sampling captures complete PK profiles, while sparse sampling enables population PK modeling with minimal patient burden.



Cox Regression

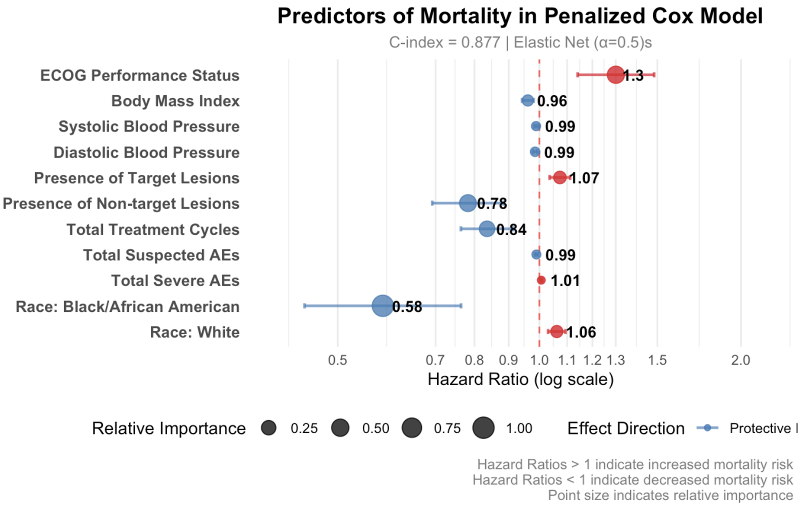


Figure: Forest Plot for Cox Regression

Random Survival Forest: Model Pipeline

Raw Data

$\downarrow$

Preprocessing (Removing Data Leakage) $\rightarrow$ Imputation $\rightarrow$ Scaling

$\downarrow$

5-Fold Cross-Validation

$\downarrow$

Data Split $\rightarrow$ Train (64%) / Val (16%) / Test (20%)

$\downarrow$

Random Survival Forest Training $\rightarrow$ 100 trees

$\downarrow$

Risk Predictions

Random Survival Forest: Model Results

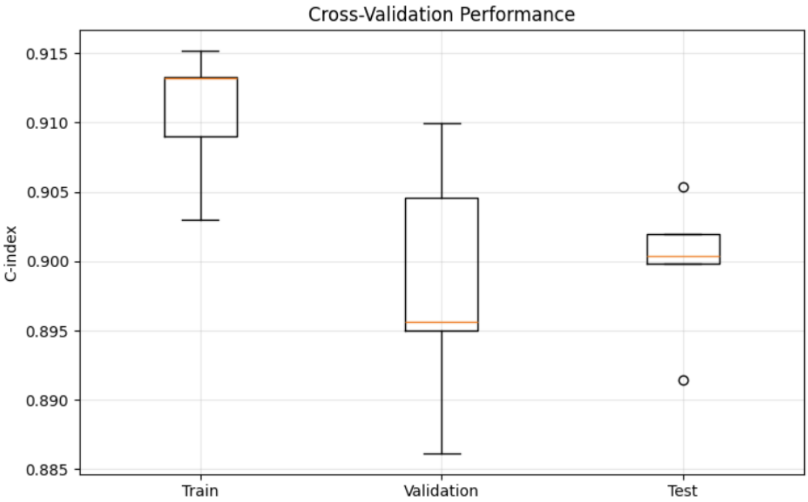


Figure: Results from Random Survival Forest

Deep Survival Network: Model Pipeline

Same Feature Engineering

$\downarrow$

Data Split $\rightarrow$ Train (80%) / Validation (20%)

$\downarrow$

Neural Network Architecture $\rightarrow$ [64, 32, 16] hidden layers

$\downarrow$

Cox Partial Likelihood Training $\rightarrow$ Negative log-likelihood loss

$\downarrow$

Risk Predictions

- **Architecture**: 3 hidden layers with BatchNorm + Dropout (20%)
- **Output**: Log-hazard ratios for survival prediction
- **Training**: Adam optimizer with learning rate 0.0005 + early stopping

Deep Survival Network: Model Results

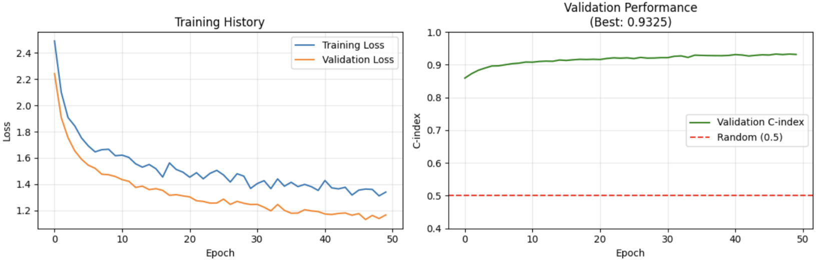


Figure: Results from Deep Survival Network

Risk Scores

- Random Survival Forest achieves the best separation

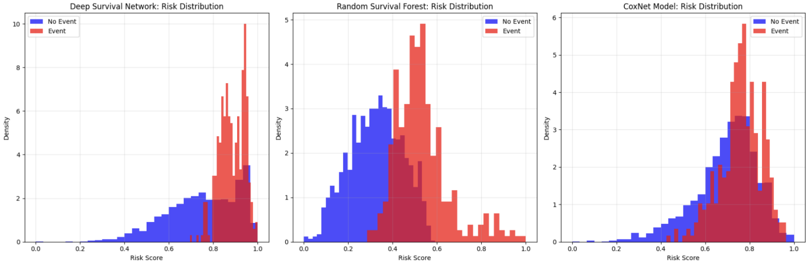


Figure: Risk Score Distribution

Important Features

1. Patient Demographics & Timing

- Patient age at baseline
- Study visit day (time progression)
- Current treatment cycle
- Total cycles so far

2. Disease Response Assessment

- Presence of target lesions
- Presence of non-target lesions

3. Adverse Event Burden

- Total number of AE events
- Count of serious AEs (Grade 3+)
- Number of body systems affected

4. Treatment Modifications & Cumulative Measures

- Number of treatment-related AEs
- Various treatment modifications (discontinuations, dose reductions)

Next Steps & Timeline

- Complete model development on main trial
- Conduct external validation on independent trial
- Perform comprehensive explainability analysis
- Finalize clinical utility assessment
- Prepare model for potential clinical implementation

Expected Outcomes

- Validated, explainable survival prediction model
- Identification of key prognostic factors
- Assessment of clinical utility
- Open-source modeling pipeline

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